Coursework or Assessment Specification

Module Details

Module Code	**UFCFVW-6-1**
Module Title	**Individual Project 3 (Course Project)**
Module Leader	Karina Kostjkina
Module Tutors	Karina Kostjkina
Year	20XX/20XX AY
Component/Element Number	PROJ3
Total number of assessments for this module	1
Weighting	**100%**
Total Assignment Time	
Element Description	Course project

Dates

Date issued to students	XX/0X/202X
Submission Date	31/05/202X
Submission Place	Moodle
Submission Time	23:55
Submission Notes	The report and application should be submitted through Moodle before the due date

Feedback

Feedback provision will be	Feedback will be provided in the Moodle environment regarding each item.

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Section 1: Overview of Assessment

The Course project is a logical continuation of the Object-Oriented Programming module and Software Engineering. This assignment is targeted at the development of the professional skills and applying knowledge received during the Software Engineering and Object-Oriented Programming modules.

This assignment assesses the following module learning outcomes:

Module Learning Outcomes	**Reference**
Understand program lifecycle and program development stages	MO1
Use user interface development	MO2
Use object-orientated paradigm for software development	MO3
Use proper coding style, considering OOP paradigm	MO4
Apply basics of documenting software products	MO5
Use of development environments and debuggers for program creation and testing purposes	MO6
Use of development environment	MO7

All work for this module module is individual and should be completed **individually**. Broadly speaking for this module, you are required to:

1. design, develop, and test an application for a given task
2. submit all source code and all repositories
3. write a technical report.

All work should be uploaded to the Moodle course which can be found at https://e.tsi.lv/course/view.php?id=2806

If you have questions about the assignment, **ask your course leader.**

Section 2: Task for programming course project

INTRODUCTION

The Course project is a logical continuation of the Software Engineering and Object-Oriented Programming modules. This assignment is targeted at the development of professional skills and applying knowledge received during the Software Engineering and Object-Oriented Programming module. Working on the Course project, you will have the ability to apply the following knowledge and skills that you acquired during the Software Engineering and Object-Oriented Programming modules:

- to develop a visual model of the system (business model, requirements model, analysis model, design model, component model, deployment model);
- to plan iterations of project;
- to estimate the quality during software project;
- to use the high level object-oriented programming language for application development;
- storing and processing information in files or in database;
- to design the graphical user interface using the graphical library of the programming environment.

1. GOAL

The main goal of the Course project is to development of an application for a specific subject area with data storage and processing using an Object-Oriented Programming Paradigm. You can choose the Object-Oriented Programming Language (for example, C#, Java, Python, C++ etc.) and type of the application (for example desktop, mobile, web etc.).

2. THE THEME OF COURSE PROJECT

Each student should complete the Course project **individually** on a topic area selected by themselves. The specific title of the Course project depends on the selected application area. An example title could be something like: *Development of the information system for library service.* Other examples of the topics can be found in the *Section 5.* Programming language, topic, title of the course project and core functionality of the application should be agreed with the teacher.

3. THE MAIN TASK

The Course Project focuses on applying fundamental principles and techniques in complex software development, emphasizing in software development processes. The primary objective is to design a visual model, implement it using object-oriented programming language, and estimate its quality. Three key deliverables include the visual model, application's source code, and a comprehensive report.

4. GENERAL REQUIREMENTS

Requirements for the Course project are presented below:

- Develop a visual model of the system (business model, requirements model, analysis model, design model, component model, deployment model).
- Select an OO language and programming environment, create an executable system code.
- Design the graphical user interface using the graphical library of the programming environment.
- Define the quality of the visual model and the executable code of the system. Evaluate the model, use the following set of metrics Chidamber and Kemerera (default) or any another known set.
- Technology development process -a unified incremental process. The number of iterations is at least three.
- Object-Oriented approach should be used for implementation.
- The program should work without critical errors.
- The program should have a clear user interface.
- The report should contain items which demonstrates the analysis, design, and development process of the application. *See Section 6 for details.*
- The report design should follow the requirements. *See Section 7 for details.*
- Data should be stored in files in any format (binary, XML, JSON etc.) or in database (MySQL, SQL Server, SQL Lite etc).

5. TASKS

1) Select a topic of your choice. *The examples of subject areas are represented in the list below*.
    1. Hotel check-in system
    2. Airline ticketing system
    3. Clinic registration system
    4. The library reader service system
    5. Travel agency customer registration system
    6. Warehouse automation system
    7. Telephone sales system
    8. Student registration system
    9. The system for registering residents of the republic
    10. Credit Card Sales System

11. The system for accounting goods in the store
12. The personnel management system of the company
13. Hotel service automation system
14. Bank customer service system
15. Beverage sales system
16. The payroll system
17. Video cassette rental system
18. The registration system for animal shelters
19. The system for registration and service of Internet clients
20. The system for registration of goods and sales in the supermarket
21. ATM service system
22. The recycling machine service system
23. Recording system for voice message reproduction
24. Vending machine control system
25. Industrial refrigerator control system
26. The washing machine control system
27. Public telephone control system
28. Metro information board management system
29. Telephone management system of an automatic telephone exchange
30. Domestic refrigerator control system
31. Suburban train turnstile control system
32. Airport Information Board Management System
33. Alarm management system
34. Game machine control system
35. Code lock control system

2) Describe functional and minimum system requirements, system users. Determine and describe the functions that the system should provide.
3) Develop a visual system model in terms of rational unified process.

   Note: For model development use any case system (for example, Visual-paradigm, CASE system Rational Rose, IBM Rational, or etc.)
4) Create and represent Business Use Case diagram (20%) in report (from business view http://www.uml-diagrams.org/use-case-diagrams.html). Describe in report all its elements (specification of actors and use case elements).
5) Create and represent System Use Case diagram in report (in terms of functionality view) http://www.uml-diagrams.org/use-case-diagrams.html). Describe in report all its elements-specification for actors and use case elements.

6) Create and represent sequence diagram for each use case element (at least one for each use case element). Show and comment all sequence diagrams in report, also represent all their scenarios. Represent initial Class Diagram. Initial quality you should represent in the table below:

Metric	Class name	Class name	Class name	Class name	Class name	Class name	Class name	Average value
WMC	value	…	…	…	…	…	…	…
NOC	value	…	…	…	…	…	…	….
System metrics								
DIT								
NC								
NOM								

7) Describe and represent in report the plan of next construction phase. Plan 3 iterations of project for supposed realization. It means that it's necessary to describe which functional possibilities of project you need to realize primarily, secondary, and thirdly.

8) Create and represent Class diagram on each iteration phase. Comment in report all its elements (classes and relationships between them).

9) Represent in report software quality (of class diagram) on each iteration. For representing use class-oriented metrics "The Chidamber and Kemerer Suite" (WMC, DIT, NOM, CBO, RFC, LCOM), see the given table below:

Metric	Class name	Class name	Class name	Class name	Class name	Class name	Class name	Average value
WMC	value	…	…	…	…	…	…	…
NOC	value	…	…	…	…	…	…	….
CBO	value							
RFC	value							
LCOM	value							
SYSTEM METRICS								
DIT								
NC								
NOM								
LOC_{Σ}								

10) Represent in report Component Diagram and describe all its elements (which classes were assigned to which components).
11) Make conclusion about course project and about the received quality assessment (is it good or bad). Files of report and visual model, project files upload to the Moodle system.

Class metrics averages at different iterations

Metric	Iteration 1 Average values	Iteration 2 Average values	Iteration 3 Average values
WMC			
NOC			
CBO			
RFC			
LCOM			
System metrics			
DIT			
NC			
NOM			
LOC_{Σ}			

12) Develop and realize a user interface.
13) Implement and test a program code.
14) Prepare the report about the completed work.

6. CONTENT OF REPORT

Below is presented the recommended structure of the report, which should be prepared as the main deliverable of the Course Project. **! Note that some parts of the report are considered in the grading scheme as separate elements.**

Title page *– use the template provided in Moodle.*

Content – *should be generated automatically and should reflect all sections and subsections of the report.*

Introduction – *this section should introduce the reader to the task you are implementing, introduces very briefly the subject area. Briefly describes each chapter (section of the report). Recommended size – max 1 page.*

1. ***Tasks for the Course project –*** *describes the tasks of the Course project in a detailed way. Recommended size – max 1 page.*
2. ***Description of the topic****– this section should include the information about the selected topic, descriptions of the requirements for the developing application, identification of the core entities objects, their attributes (characteristics), the interrelation between objects. We are also welcome here not only text but figures, tables, diagrams etc.*
3. ***Design*** *– this section should include information about the visual model, design of the application. Provide here your vision of how navigation will be organised (menu design), what user types you will use to describe objects. Provide information about files that will store the data (Data should be stored in files in any formats (binary, XML, JSON etc) or in database (MySQL, SQL Server, SQL Lite etc). Itš possible to present here ERD or file structure (binary, XML, JSON etc).*

   Present here use case, sequence, class and deployment diagrams, algorithms as flowcharts for the methods you are planning to implement, forma or pages layouts. We are also welcome here not only text but figures, tables, diagrams etc.
4. ***Development*** *– provide in this section information about the development. Describe in detail, created user-defined types, developed user interface, list of source files. We are recommending utilising tables. For example, a class could be described in the following way: fields with a type and a description, properties with a type and a description, methods with a signature, description of the result and of the arguments. provide here information about the developed application and its functionality. Demonstrate screenshots of the developed applications and results of completing different operations. Represent program code interesting examples.*
5. ***Testing*** *– provide here information about how you have organised the testing of the developed application. List here scripts, scenarios, test cases, test results, details about found errors and comments, how errors have been corrected or not corrected.*
7. ***Conclusions*** *– Indicate in this section what has been done in the frame of the course project. Sum up information about the application, development process, and* ***quality****. Briefly describe the main functionality*

of the application. Indicate the main issues during the development process. Complete the analysis of your application, what has been done well, what could be improved, what has caused most of the problems during design, development, testing etc.

8. References *– any references you have used.*

Annexe: *source code.*

7. FORMATTING YOUR REPORT

Report design requirements that should be considered preparing the report:

- **Pages:**
  - **Page size:** A4
  - **Margins** - left - 3 cm; top, bottom - 2 cm, right-1,5
  - **Page numbering** – at the bottom of the page in the middle, the title page should not be enumerated.
- **Heading**:
  - 1st level - 14 pt, Times New Roman, Bold, Upper Case, alignment: left (except Introduction, conclusion, reference list, these headings must be centered).
  - 2nd level – 12pt, Times New Roman, Bold, Sentence Case, alignment: left
  - 3rd level – 12pt, Times New Roman, Italic, Sentence Case, alignment: left
- **Main text:**
  - Font Times New Roman, size 12pt, line spacing 1.5.
  - First-line indent **-** 1 cm.
  - Alignment – justify.
- **Figures/Pictures:** alignment: center
  - **Caption template**: Fig. <number> Title of the figure,
  - **Caption text:** Font Times New Roman, size 10pt
  - **Caption location:** Under figure, centred
- **Tables**: alignment: center
  - **Caption template**: Table <number> Title of the table
  - **Caption text**: Font Times New Roman, size 10pt, linespacing-single
  - **Caption location**: Above the table, centred
- **Sections** – each section (chapter) starts on a new page.
- **Reference** – if any, we are recommending using Harvard style.
- **Application (code)** – font-size: 8 font size, line spacing – 1

8. GRADING SCALE

We are strongly recommending making yourself familiar with the **grading scheme** presented below.

An application, files with source code and a report submission required!

Maximum number of points -**100 points**
Passing points amount – **40 points.**

	Criteria	Marks						
	Report design	10	The outstanding design of the report. All requirements are fulfilled, the report has a uniform design. 10 points	The design of the report follows the requirements, but some minor issues are observed. 8 points	The report in general follows the requirements but has a significant number of minor issues in the design. 6 points	The report attempts to follow the design requirements. But major issues are observed, they make report partly ununiformly designed. 4 points	The report is designed in ununiform style, but it does not follow the requirements. 2 points	The report does not follow the design requirements at all. It has an ununiform design. Or the report is not submitted. 0 points
	Topic and problem area analysis and description. **Inception Stage.** Functional & System Requirements. System users Initial Requirements model.	15	Outstanding description of the topic. The analysis of the topic is completed, results are clear and well presented. Requirements are fully covered. Amazing presentation and description of Functional & System Requirements, System users. 15 points	The description is clear and covers all the required points. But some extension could be done. Functional & System Requirements, System users were defined partly and well presented. 12 points	A moderate description is provided. It gives a view on the topic. Functional & System Requirements, System users were defined partly and moderately presented. 9 points	The description is provided in general form, but at the same time attempts to complete the analysis is foreseen. Functional or System Requirements or System users were defined partly and poorly presented. 6 points	The description is presented in a minimal volume. It is too generic and does not provide a clear vision on the topic. No attempts to complete the analysis. Functional or System Requirements or System users were very poorly defined. 3 points	The description is not presented, or the description does not provide a clear vision on the topic. Functional or System Requirements or System users were not defined. 0 points

	Criteria	Marks						
	Design section of the report. **Elaboration Stage.** Final Requirements model. Scenarios for Use Case Elements.	15	Outstanding presentation of the design section of the report. The provided information is clear and does not require additional comments/extension /clarification. The section has an excellent relationship with the section Analysis of the subject area. Amazing presentation and description of Requirements model, Scenarios for Use Case Elements. 15 points	The report has a section devoted to the design. The provided information is clear has a good relationship with the information provided in section Analysis of the subject area. All requirements are well covered, only minor issues are identified which requires additional clarification. Requirements model, Scenarios for Use Case Elements well presented. 12 points	The report part devoted to the design is presented in the report. The presented information is moderate, but it the same time it gives useful and understandable information about the future application. There is a significant number of minor issues, which could be extended/clarified/described to make this section clear. Moderate relation with the section Analysis of the subject area is observed. Requirements model, Scenarios for Use Case Elements moderately presented. 9 points	The report part devoted to the design is presented in the report. Provided information gives a general vision on the design of the application. The information corresponds in general to the requirements, but the significant number of gaps in the description does not allow to get a clear understanding of the information. Moderate relation with the section Analysis of the subject area is observed. Requirements model, Scenarios for Use Case Elements poorly presented. 6 points	The report has a section devoted to the design. But provided information is limited or partly covers requirements. It does not provide a clear vision on the design of the application. Provided information has a partial relationship with the section Analysis of the subject area. Requirements model, Scenarios for Use Case Elements partly and very poorly presented. 3 points	The report does not include information about application design. The section is missing or provided information is out of scope. Requirements model, Scenarios for Use Case Elements not presented. 0 points
	Design section of the report. **Elaboration Stage.** Interaction Diagrams. Initial Class Diagram. Initial Quality. Plan for Construction stage	15	Outstanding presentation of the development section of the report. The provided information is clear and does not require additional comments/extension /clarification. Amazing presentation and description of Interaction Diagrams, Initial Class Diagram, Initial Quality, Plan for Construction stage. 15 points	The report has a section devoted to development. All requirements are well covered, only minor issues are identified which requires additional clarification. Interaction Diagrams, Initial Class Diagram, Initial Quality, Plan for Construction stage are well presented. 12 points	The report part devoted to the development is presented in the report. The presented information is moderate, but it the same time it gives useful and understandable information about the development. There is a significant number of minor issues, which could be extended/clarified/described to make this section clear. Interaction Diagrams, Initial Class Diagram, Initial Quality, Plan for Construction stage are moderately presented. 9 points	The report part devoted to the development is presented in the report. Provided information gives a general vision on the development of the application. The information corresponds in general to the requirements, but the significant number of gaps in the description does not allow to get a clear understanding of the information. Interaction Diagrams, Initial Class Diagram, Initial Quality, Plan for Construction stage are poorly presented. 6 points	The report has a section devoted to development. But provided information is limited or partly covers requirements. It does not provide a clear vision on development. Interaction Diagrams, Initial Class Diagram, Initial Quality, Plan for Construction stage are very poorly presented. 3 points	The section is missing or provided information is out of the scope. Interaction Diagrams, Initial Class Diagram, Initial Quality, Plan for Construction stage are not presented 0 points

	Criteria	Marks						
	The development section of the report. **Construction Stage**. Realization of 1-3 Iterations/ Logical Model. Quality Estimation	15	Outstanding presentation of the development section of the report. The provided information is clear and does not require additional comments/extension /clarification. Amazing presentation for Realization of 1-3 Iterations, Logical Model, Detailed Quality Estimation. 15 points	The report has a section devoted to development. All requirements are well covered, only minor issues are identified which requires additional clarification. Realization of 1-3 Iterations, Logical Model, Quality Estimation are well presented. 12 points	The report part devoted to the development is presented in the report. The presented information is moderate, but it the same time it gives useful and understandable information about the development. There is a significant number of minor issues, which could be extended/clarified/described to make this section clear. Realization of 1-3 Iterations, Logical Model, Quality Estimation are presented moderately. 9 points	The report part devoted to the development is presented in the report. Provided information gives a general vision on the development of the application. The information corresponds in general to the requirements, but the significant number of gaps in the description does not allow to get a clear understanding of the information. Realization of 1-3 Iterations, Logical Model, Quality Estimation are poorly or partly presented. 6 points	The report has a section devoted to development. But provided information is limited or partly covers requirements. It does not provide a clear vision on development. Realization of 1-3 Iterations, Logical Model, Quality Estimation are very poorly or partly presented. 3 points	The section is missing or provided information is out of the scope. Realization of 1-3 Iterations, Logical Model, Quality Estimation are not presented. 0 points
	Design section of the report **System Architecture.** Component Diagram	5	Outstanding presentation of the development section of the report. The provided information is clear and does not require additional comments/extension /clarification. Amazing presentation of the Architecture`s system. 5 points	The report has a section devoted to development. All requirements are well covered, only minor issues are identified which requires additional clarification. Architecture of the system is well presented. 4 points	The report part devoted to the development is presented in the report. The presented information is moderate, but it the same time it gives useful and understandable information about the development. There is a significant number of minor issues, which could be extended/clarified/described to make this section clear. Architecture of the system is presented moderately. 3 points	The report part devoted to the development is presented in the report. Provided information gives a general vision on the development of the application. The information corresponds in general to the requirements, but the significant number of gaps in the description does not allow to get a clear understanding of the information. Architecture of the system is poorly or presented partly. 2 points	The report has a section devoted to development. But provided information is limited or partly covers requirements. It does not provide a clear vision on development. Architecture of the system is very poorly or partly presented.presented. 1 points	The section is missing or provided information is out of the scope. Architecture of the system is not presented. 0 points

	Criteria	Marks						
	Development results section of the report. **UI, program code**.	20	Outstanding presentation of the development results section of the report. The provided information is clear and does not require additional comments/extension /clarification. 20 points	The report has a section devoted to the development results. All requirements are well covered, only minor issues are identified which requires additional clarification. 16 points	The report part devoted to the development results is presented in the report. The presented information is moderate, but it the same time it gives useful and understandable information about development results. There is a significant number of minor issues, which could be extended/clarified/described to make this section clear. 12 points	The report part devoted to the development results is presented in the report. Provided information gives a general vision on the development results of the application. The information corresponds in general to the requirements, but the significant number of gaps in the description does not allow to get a clear understanding of the information. 8 points	The report has a section devoted to the development results. But provided information is limited or partly covers requirements. It does not provide a clear vision on development results. 4 points	The section is missing or provided information is out of the scope. 0 points
	Conclusion about Quality and conc. about project	5	Outstanding conclusion. They sum up information about the developed application and development process. They demonstrate analysis results and emphasise the issues. Provides critics about development selected solutions etc. 5 points	Good conclusions, which sum up information about the developed application and development process. The attempt to complete the analysis and present analysis results at a good level. 4 points	Moderate conclusions. They demonstrate what has been done in details. Moderate attempt to make analysis and indicate any issues during development. 3 points	Most of the conclusions are too general. Conclusions present what has been done in a generic form. No attempt to make an analysis and indicate any issues during development. 2 points	Conclusions are very light. They are not clear, do not demonstrate what has been done in the frame of the project. No attempt to make an analysis and indicate any issues during development. 1 point	The section is missing or provided information is out of the scope. 0 points